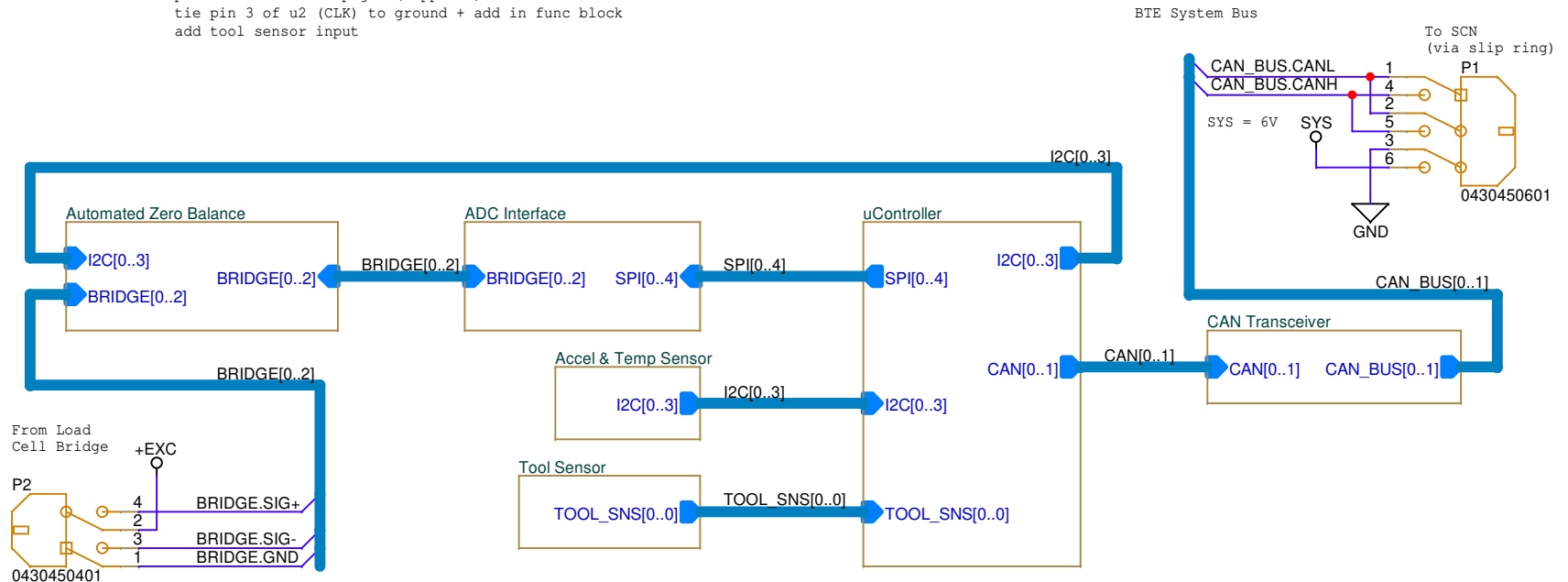


This Primus subsystem is the Torque Interface Board.
The primary functions of the HSA are to:
Read the load cell signals.

2nd Prototype Todos: 2nd Prototype Completed Todos:
Fix U2 - Pin 3 needs to be tried to ground (and created)
upgrade uC to same footprint but m4 - STM32L433CC
upgrade crystal speed? 50MHz? no all clocks 20M
consider changing zero-balance resistors to narrow
the adjustment range.
need BOOT0 pin on uC PH3
put all of uC on 1 page? (supplies)
tie pin 3 of u2 (CLK) to ground + add in func block
add tool sensor input

REVISION HISTORY

REV	DESCRIPTION	DATE	APPROVED	ECO
A	Prototype	10/15/2024	JPN	NA



Prototype 2 Todos;

Hold-up & Regulation

PROPRIETARY STATEMENT

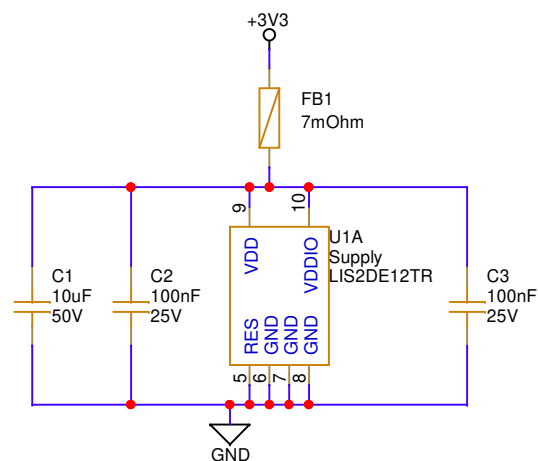
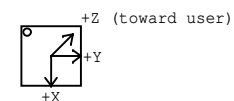
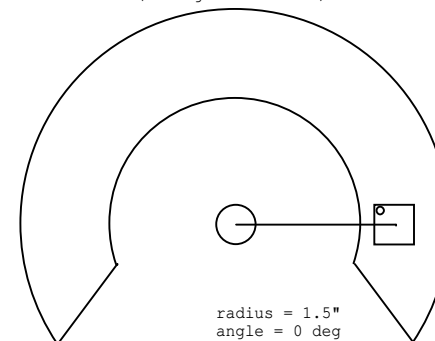
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Title: Torque Interface Board			
Size: A	Document Number: 20140002	Page Name: Torque Interface Board	Rev: A
Filename (.dsn): TORQUE INTERFACE BOARD			
Last Edited By: J.Normand			
Date: Tuesday, October 15, 2024 15:50:54			Sheet 1 of 9

LISDE12 on Top
(facing toward user)



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Torque Interface Board

Size:
A

Document Number:
20140002

Page Name:
Accel & Temp Sensor

Rev:	A
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Filename (.dsn): TORQUE INTERFACE BOARD

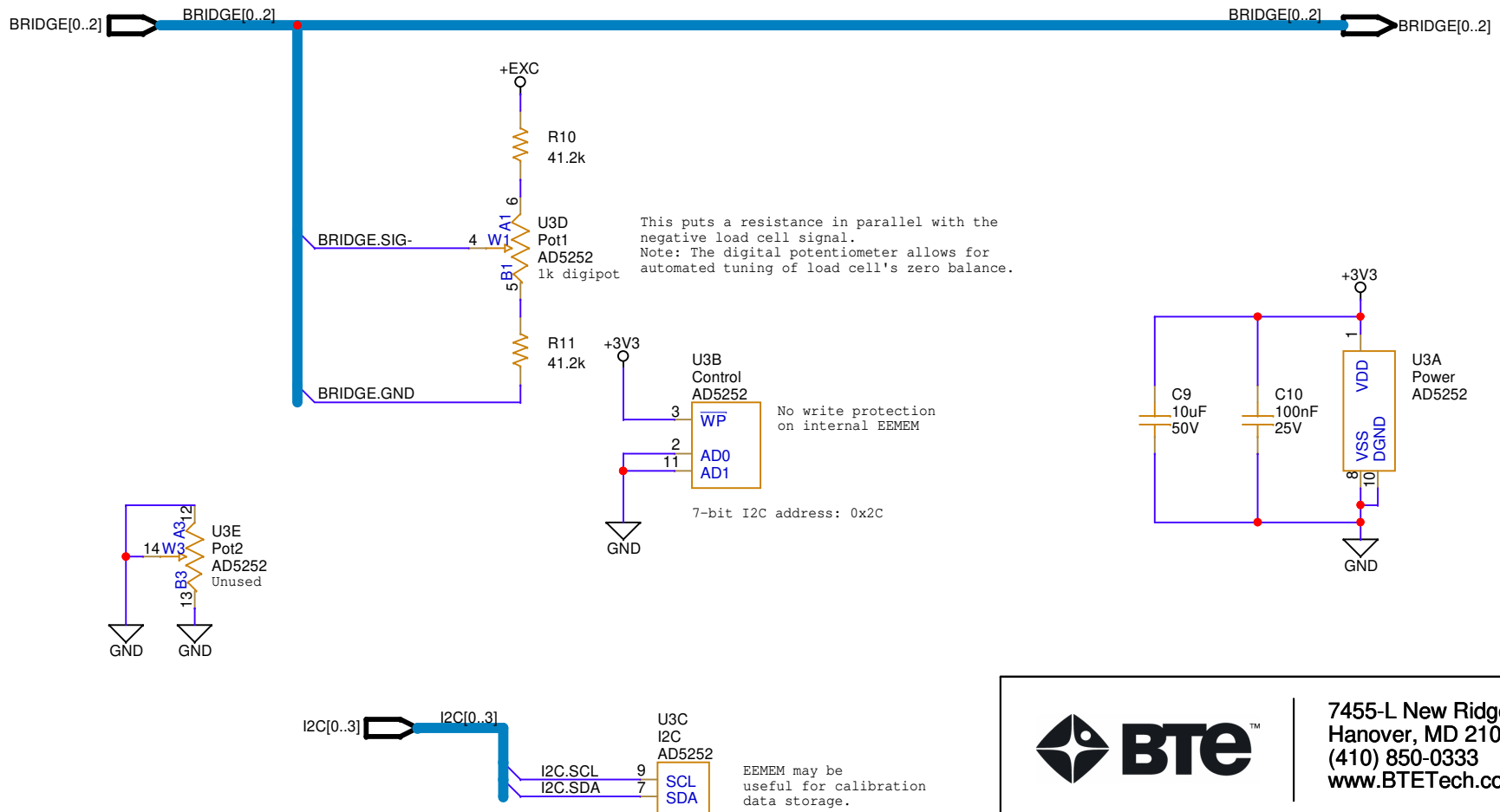
Last Edited By: J.Normand

Date: Tuesday, October 15, 2024 15:50:54

Sheet 2 of 9

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This HSA subsystem automates the zero balancing of the load cell.
The digital potentiometer is controllable via I2C.
The digital potentiometer alters the negative signal resistances.

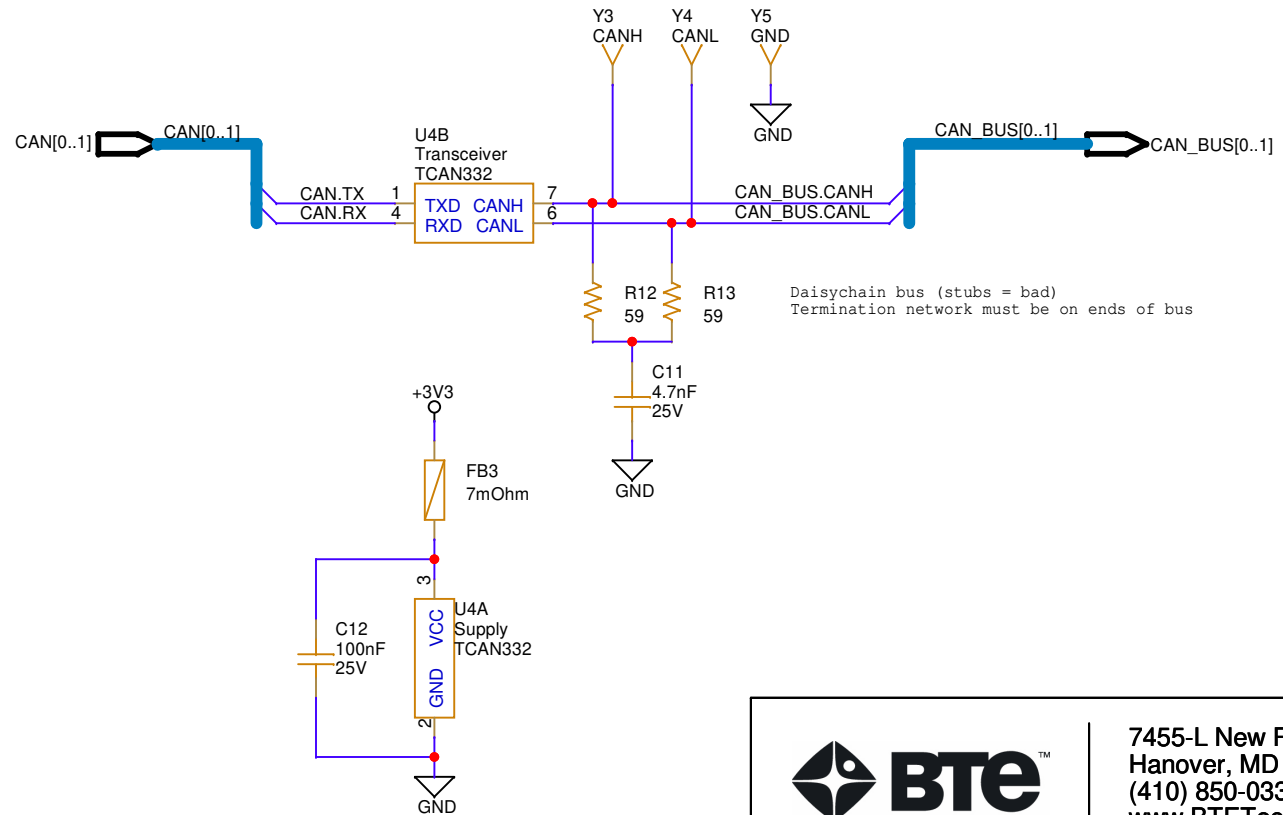


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This HSA subsystem communicates with the BTE Sytem Bus.
 The BTE System Bus uses CAN hardware layer.
 The BTE System Bus uses CANopen for the protocol layers.
 The BTE System Bus uses a producer/consumer protocol.



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Size: A	Document Number: 20140002	Page Name: CAN Transceiver	Rev: A
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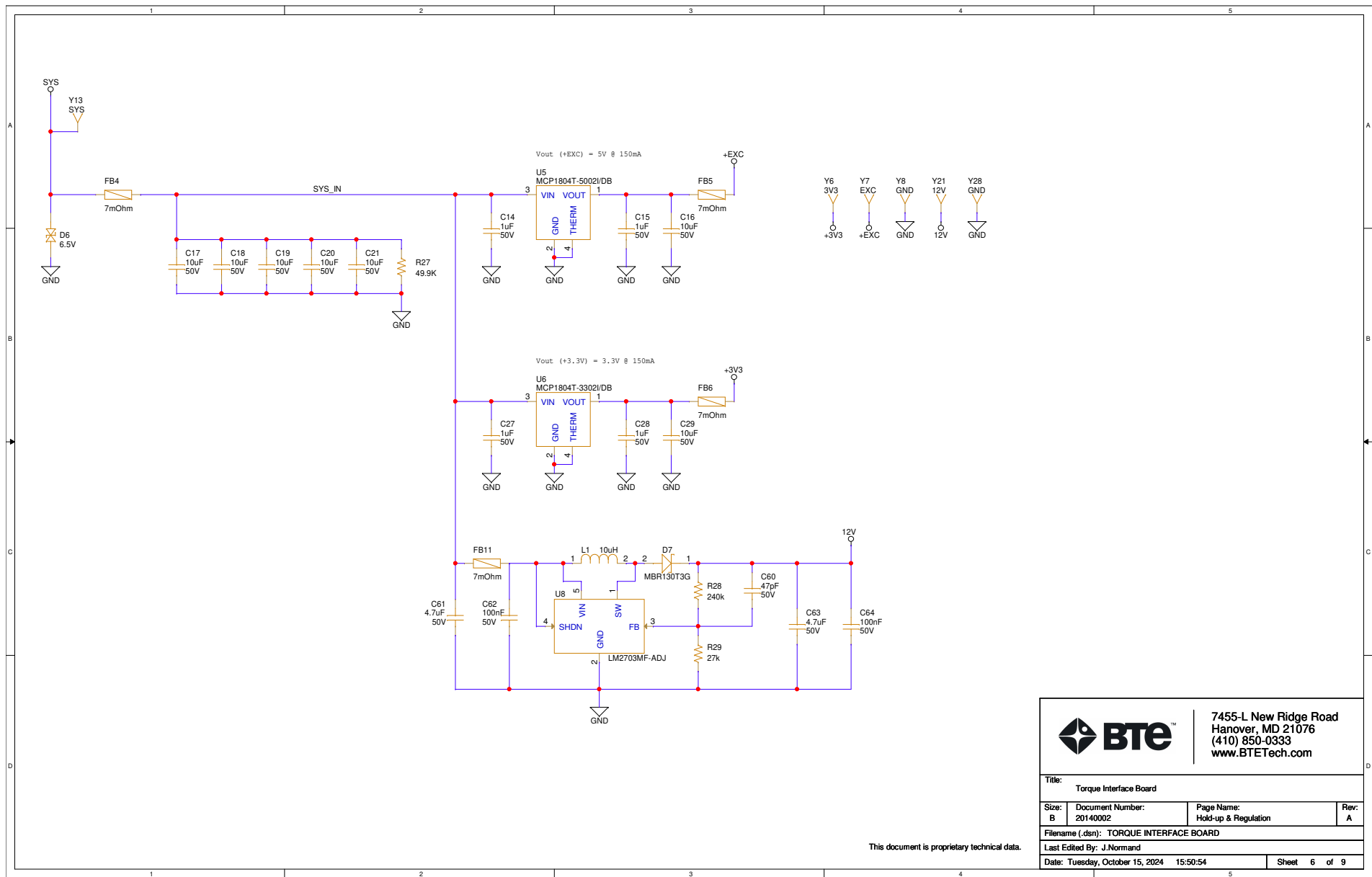
Filename (.dsn): TORQUE INTERFACE BOARD

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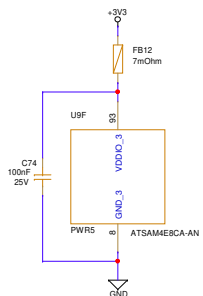


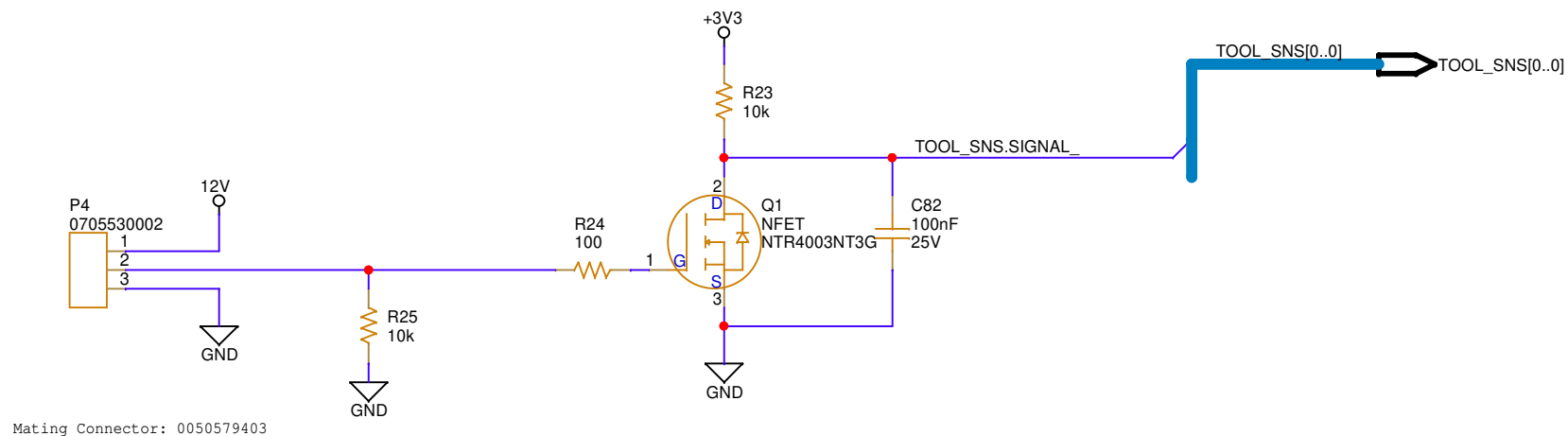
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Title: Torque Interface Board			
Size: B	Document Number: 20140002	Page Name: Hold-up & Regulation	Rev: A
Filename (.dsn): TORQUE INTERFACE BOARD			
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